

Qatar-2b

R-1589

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_220206 · created 2026-08-02T00:49:06+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459616.8952 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.051 +/- 0.065
Transit depth (Rp/Rs)^2	0.26 +/- 0.66 [%]
Orbital Inclination (inc)	85.12 +/- 1.75
Airmass coefficient 1 (a1)	0.66 +/- 0.4
Airmass coefficient 2 (a2)	0.19 +/- 0.12
Scatter in the residuals of the lightcurve fit is	64.2 %
Best Comparison Star	#3 - [418, 142]
Optimal Aperture	3.21
Optimal Annulus	7.31
Transit Duration (day)	0.059 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.051 $\pm$ 0.065 vs 0.16327 (deviation 1.73 σ)
Duration fit vs published	0.059 d vs 0.0767 d (deviation 1.61 σ)
Mid-time O-C	-2.7 min
Depth SNR	0.8
Residual scatter	64.2 %

Light curve

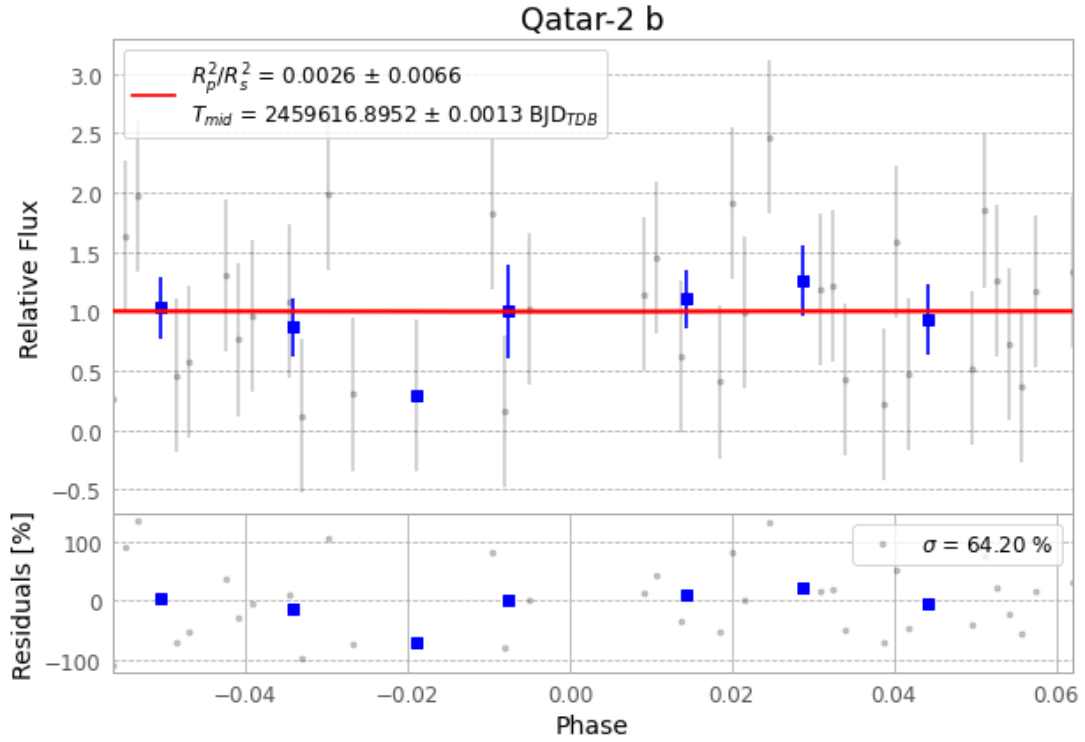


Figure 1 — FinalLightCurve_Qatar-2 b_2022-02-06.png (SHA-256 287c8dcd39358062...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [20, 160], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 ce4da406bcde9eb9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

75 FITS frames (49 MB), Qatar-2220206073612.FITS → Qatar-2220206112412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-220206.log (4378a8b65f72...), FinalLightCurve_Qatar-2 b_2022-02-06.png (287c8dcd3935...), FinalLightCurve_Qatar-2 b_2022-02-06.csv (544cb2ea95e4...)

Compute resources

Wall time 215.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 00:49Z.